The Silent Infiltrators: A Survey of Software-Based Side-Channel Attacks and Their Defenses in SDNs

Unveiling the Hidden Attack Vectors in Modern Networking

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Abstract

Software-defined networks (SDNs) have surfaced as an innovative paradigm in networking, promising heightened adaptability and control. However, with progress comes vulnerabilities; they're not immune to cyber risks. Notably, software-driven side-channel attacks have become a major concern due to their covert operations and potential harm. This survey presents an all-encompassing examination of such clandestine incursions, encompassing everything from covert enumeration to side-channel intrusive methods. This study offers insights into the intricacies and ramifications of these weak points. Addressing these issues, we delve into both anticipatory and responsive security strategies, assessing their viability, expandability, and operational implications. The survey concludes by looking ahead,identifying ongoing research gaps, and pondering the contributions of advancements like machine learning to bolster SDN controls.

Introduction

Software Defined Networks (SDNs) have transformed computer networks by offering enhanced scalability, robustness, and manageability.

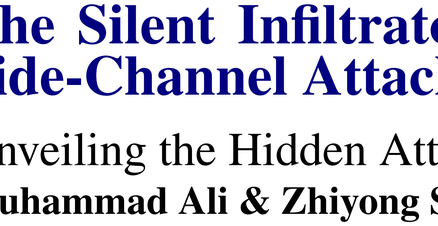


Figure 1: Software Defined Networking Market Size — *Source: Modor Intelligence*

Software-based side-channel attacks (SCA) in SDNs exploit subtle data leakages, targeting the software layer to infer sensitive network information.

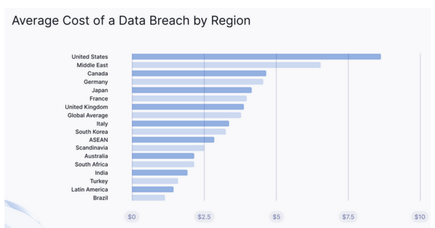


Figure 2: Cost of Data Breach Report 2020 — *Source: IBM & Ponemon*

Main Objectives

1. Providing an overview of software-based side-channel attacks in SDNs.
2. Analyzing inherent vulnerabilities in SDN implementations.
3. Elucidating various attack techniques and their impacts.
4. Evaluating existing mitigation strategies.

SDN Overview

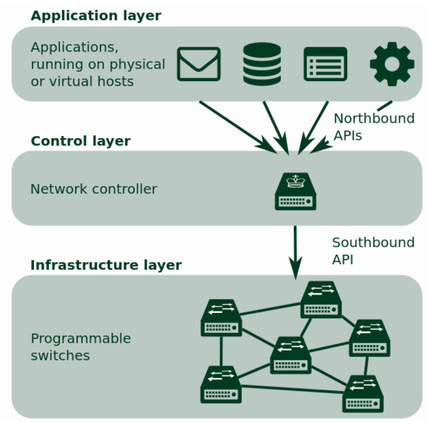


Figure 3: SDN architecture concept

SDN Security

Risks of Centralized Control in SDNs

- ✱ Single Point of Failure
- ✱ Complex Security Management

Understanding Side Channel Attacks (SCA)

- ☛ Logical SCA are a type of security exploit that take advantage of the information inadvertently leaked by software.
  - Non-Intrusive Nature
  - Stealthy
  - Exploiting Statistical Methods

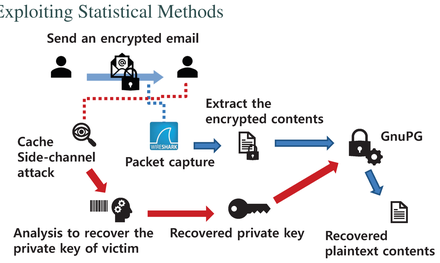


Figure 4: Overview of SCA scenario

Two Main Categories of Side-Channel Attacks in SDN

❶ Covert Channel Communication Attacks — Teleportation Exploits
❷ Information Gathering Attacks — Reconnaissance Attack

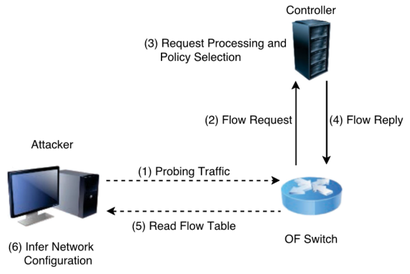


Figure 5: Overview of SCA Reconnaissance Attack

Key Attack Surfaces

- ➳ Centralized Control
- ➳ Programmability
- ➳ Open Interfaces and APIs

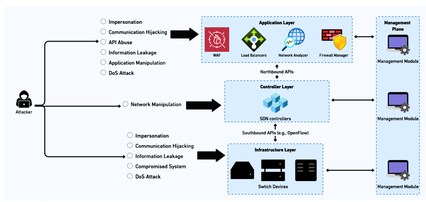


Figure 6: Cyber-Attacks and Attack Surfaces in SDN

Threat Model

Attack Method	Risk	Complexity
Accessing Flow Tables via Passive Listening Ports	High	Medium
Exploiting Poor Device Configurations	High	Low
Using RTT Variance for Information Inference	Medium	High
Sniffing Control Traffic	High	Medium
Exploiting Built-in Backdoors	Very High	Medium
Compromising the Switch	Very High	High

Table 1: Comparing Attack Methods with Associated Risk and Complexity Levels

Countermeasures

Countermeasure	False Positives	Performance
Flow Obfuscation	Low	Medium
Rate Limiting	Variable	Low
Encryption	N/A	High

Table 2: Comparative Analysis of SDN Countermeasures

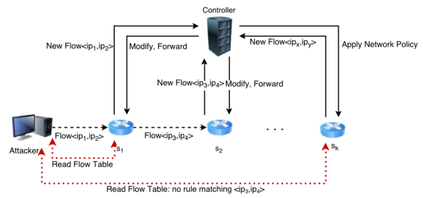


Figure 7: Overview of the flow obfuscation countermeasure — The dotted lines indicate the flow table side-channel.

Conclusions

- Central point of control, if compromised, can lead to widespread network issues.
- Attackers exploit the necessary, albeit sporadic, communication between switches and the controller to gather sensitive information.
- To prevent these attacks, measures can be implemented to ensure that the actions of one host do not influence the operation time of another host.

Forthcoming Research

Two main areas for future research are identified:

■ Machine Learning in Cybersecurity
  ✎ Investigating how ML can detect these patterns and how attackers might evade such detection is a key research focus

■ Anti-Header Spoofing
  ✎ Exploring methods to prevent ID (MAC address, DIDP, IP addresses) or header spoofing could shut down or complicate many attack strategies

References

[1] *Flow Reconnaissance via Timing Attacks on SDN Switches*, 2017.

[2] A novel stealthy attack to gather sdn configuration-information. *IEEE Transactions on Emerging Topics in Computing*, 2020.